

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

ES-220(CE)-CBCS

B.E. III Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Material Science

Time : Three Hours

Maximum Marks : 60

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain in detail about Composite materials and its Uses?
b) What is Terra cotta? How it is manufactured?
2. a) Iron is basic construction material explain and write importance of steel in different form in construction work.
b) Write and discuss the name of various non ferrous metals and their applications in construction work.
3. a) Define cement mortar, its functions and their quality.
b) Write in brief on:
 - i) Testing of concrete
 - ii) Bulking of sand
 - iii) Lime mortar
4. What do you understand by “Bitumen” and “Tar”? Also give their applications in civil engineering works.

ES-220(CE)-CBCS

PTO

[2]

5. a) Draw the stress-strain curve for mild steel and show the silent points.
b) Difference between edge and screw dislocation.
6. Explain the following :
i) Recovery
ii) Recrystallization
iii) Grain growth

OR

Define “Fatigue” and “Creep”. Also explain corrosive resistant property of materials.

7. What do you understand by “Curing of concrete”? Also explain ductility property of steel.

OR

- a) Explain the constituents of a bituminous mix.
b) Explain in detail flexibility and brittleness.

8. Write short note on (any two)
i) Nano materials
ii) Fracture
iii) Hardened concrete
vi) Ceramic tiles
